

Training Configuration and Summary:

Dataset used: Training: rssi_odom_dataset_12apr25_isolado.parquet | Experiment: rssi_odom_dataset_11apr25_isolado.parquet

model_type: LSTM
group_sec: 100
timesteps_orig: 60
batch_size: 256
min_units: 16
max_units: 256
units_step: 16
dropout_min: 0.2
dropout_max: 0.3
dropout_step: 0.1
activation_functions: ['tanh', 'relu', 'elu']
optimizers: ['adam', 'rmsprop', 'Nadam']
min_layers: 1
max_layers: 2
max_epochs: 50
hyperband_iterations: 3
group_msec: 100
samples_per_sec: 10
forecast_horizon_sec: 4
forecast_steps: 40
timesteps: 60

Dataset Parameters:

Training: $\Delta t \approx 12000.0$ ms | samples/window=1.00 | timesteps=60
Experiment: $\Delta t \approx 12000.0$ ms | samples/window=1.00 | timesteps=60

Tuning:

Trials executed: 10/10

Training Performance:

Loss: 0.0061 | MAE: 0.0597

Validation Performance:

Loss: 0.0094 | MAE: 0.0781

LSTM Multi-Step Training Report

Report Generated: 2025-09-14 13:39:17

Training Start: 2025-08-05_08-33-05

Training End: 2025-08-05_08-36-44

Training Duration: 3.65 minutes

Predictions Start: 2025-08-05_08-36-44

Predictions End: 2025-08-05_08-41-28

Predictions Duration: 4.73 minutes

Best Hyperparameters:

Number of Layers: 1

Activation: tanh

Optimizer: adam

Dropout Rate: 0.20

MAE: 1.4546

MSE: 3.6907

RMSE: 1.9211

R2 Score: 0.8582

Multi-Step Forecast:

Forecast Steps: 40

MAE: 0.0526

MSE: 0.0046

RMSE: 0.0679

R2 Score: 0.9027

Dataset size before down-sample: 40,318 rows
Dataset size after down-sample: 12,116 rows

First 5 timestamps BEFORE:

1744397593300
1744397593310
1744397593332
1744397593362
1744397593375

...

Last 5 timestamps BEFORE:

1744398077967
1744398077978
1744398077991
1744398078003
1744398078016

First 5 timestamps AFTER:

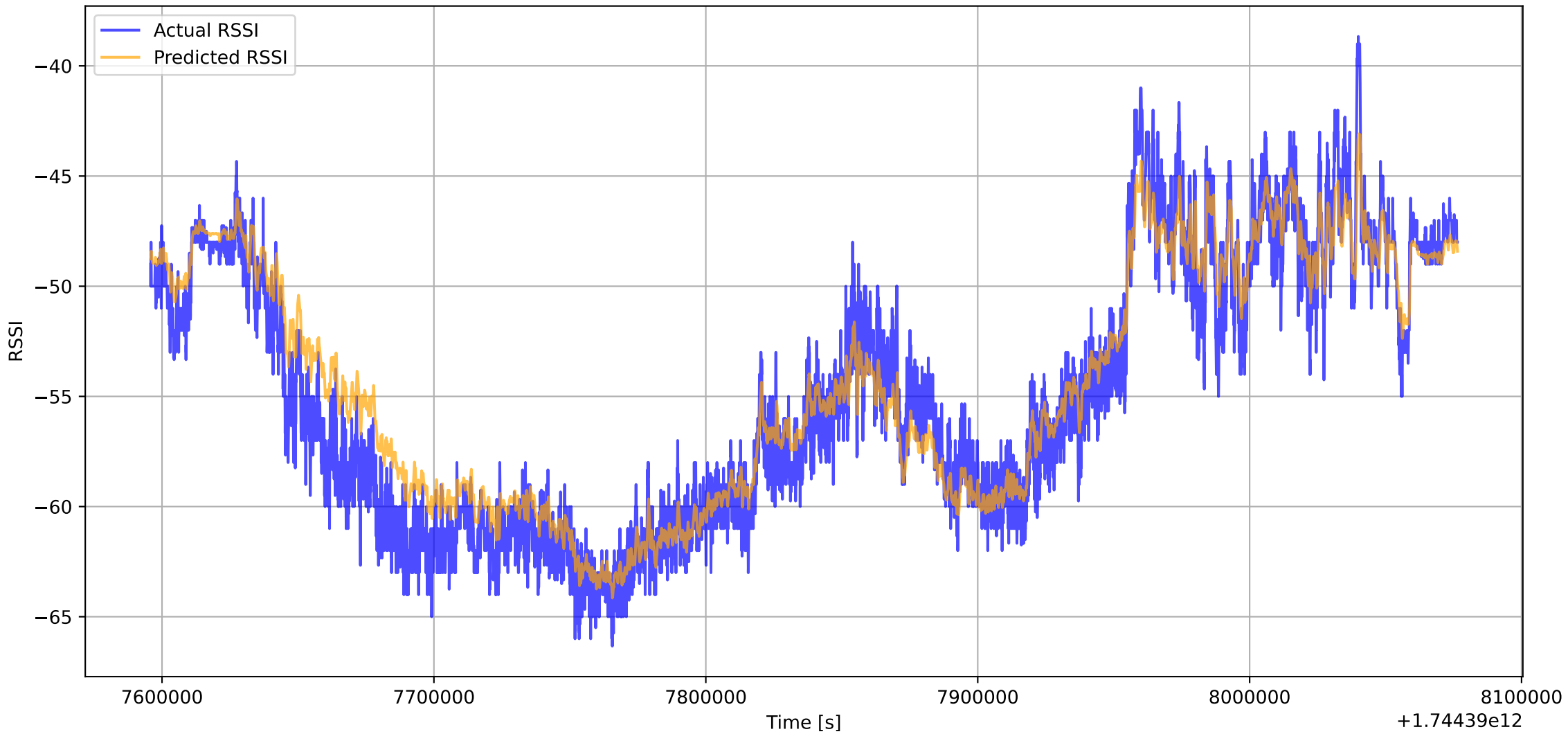
1744397593300
1744397593332
1744397593362
1744397593536
1744397593568

...

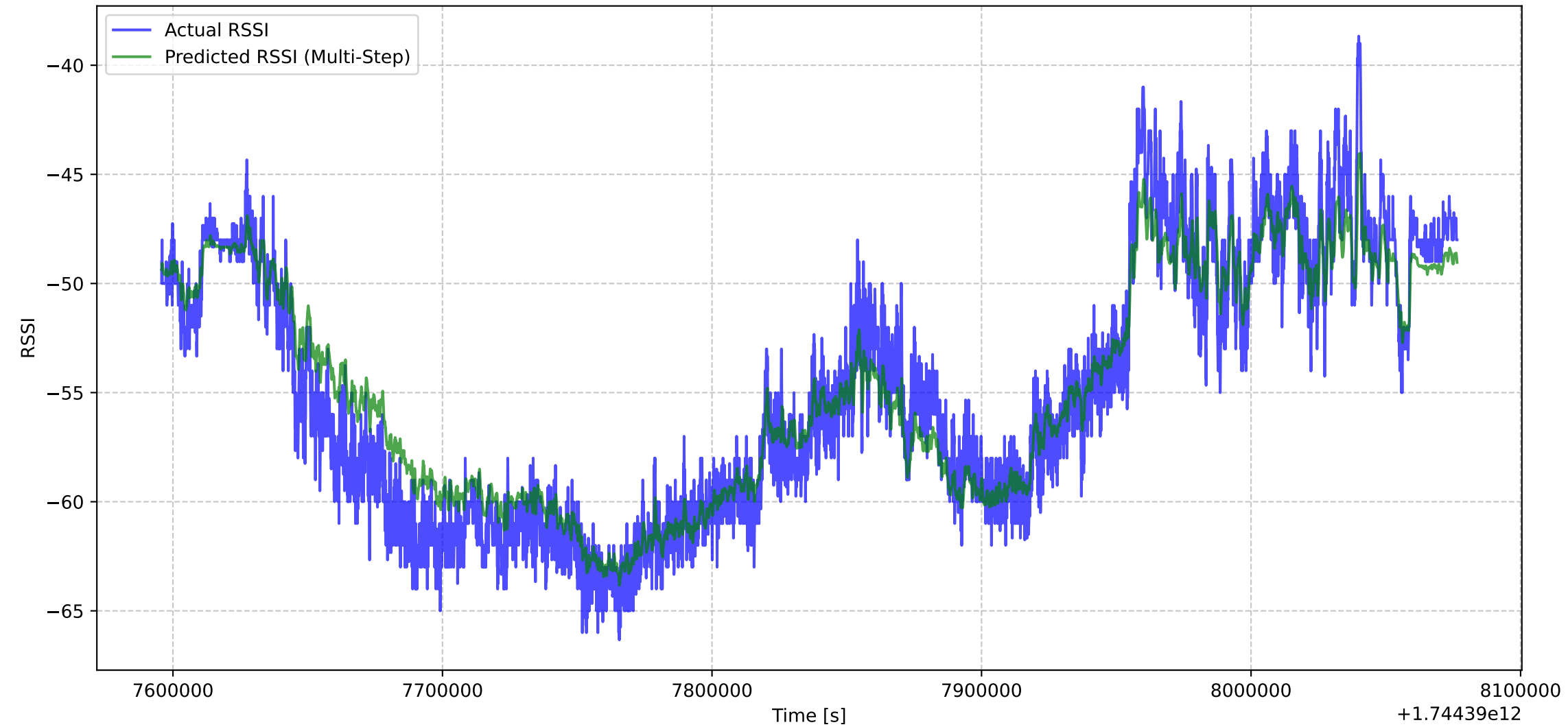
Last 5 timestamps AFTER:

1744398077847
1744398077883
1744398077921
1744398077967
1744398078003

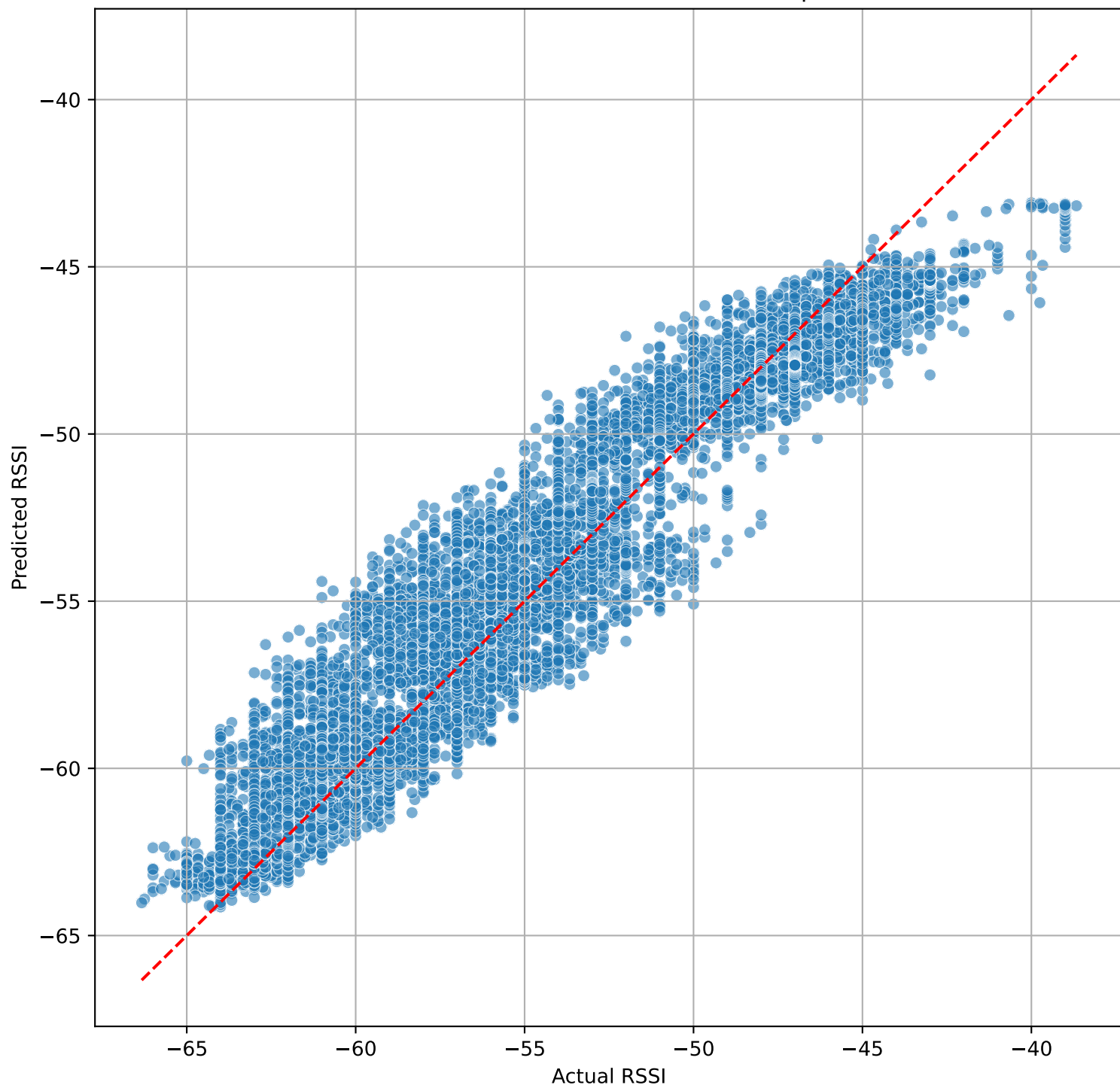
Actual vs Predicted RSSI (First Step)



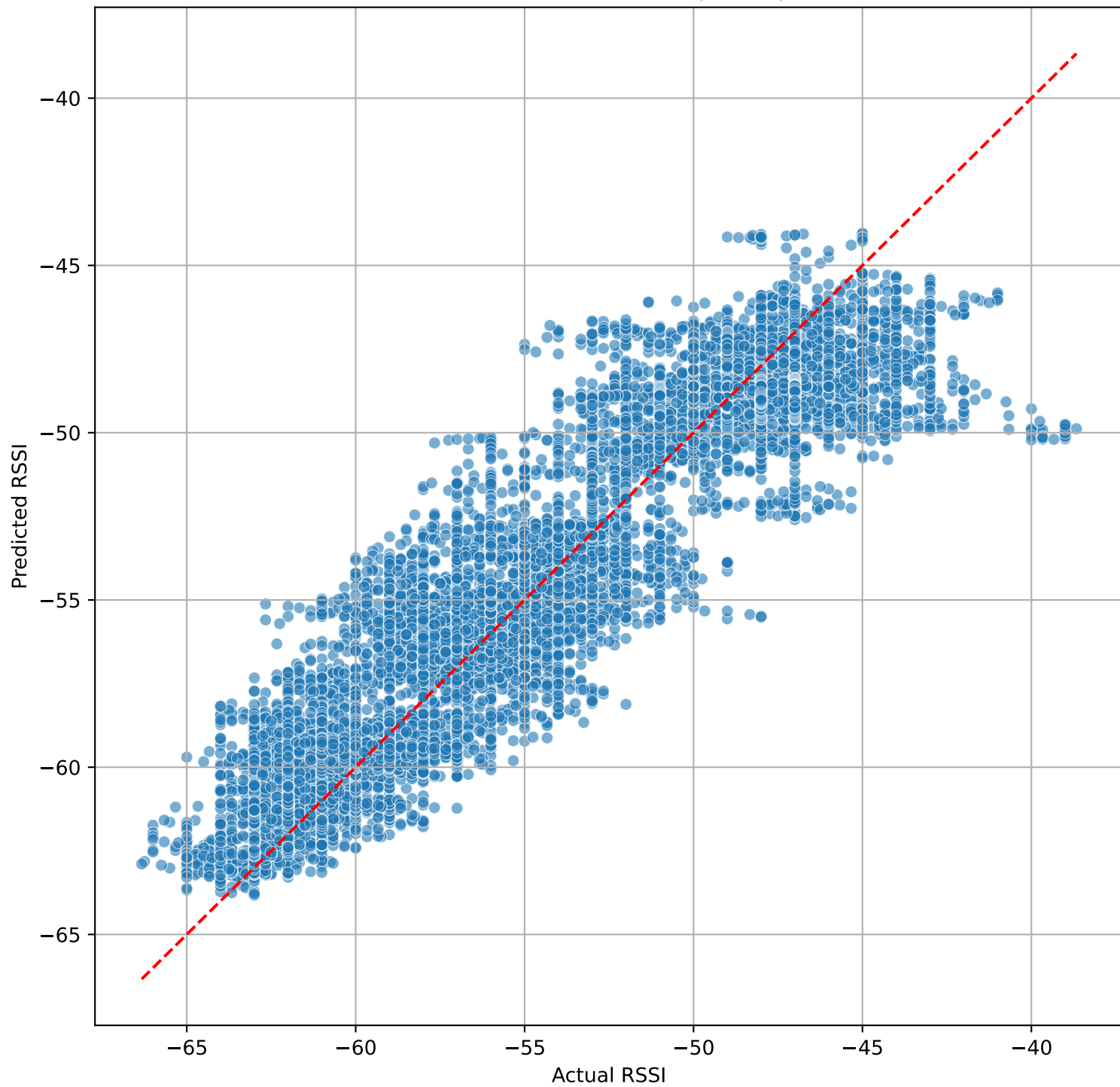
Actual vs Predicted RSSI (Multi-Step (step 40 $\approx$ 4.0 s))



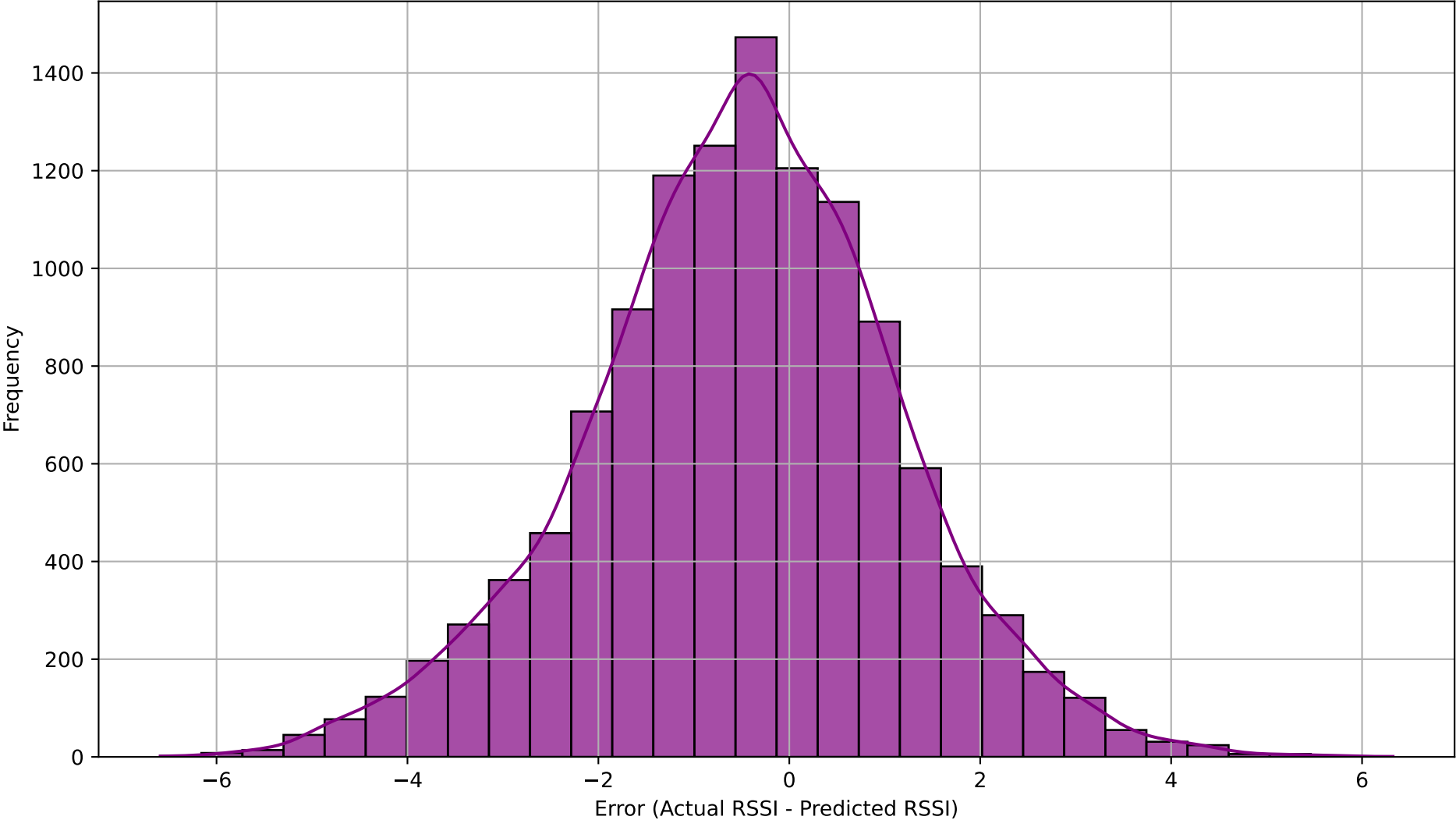
Actual vs Predicted RSSI (First Step)



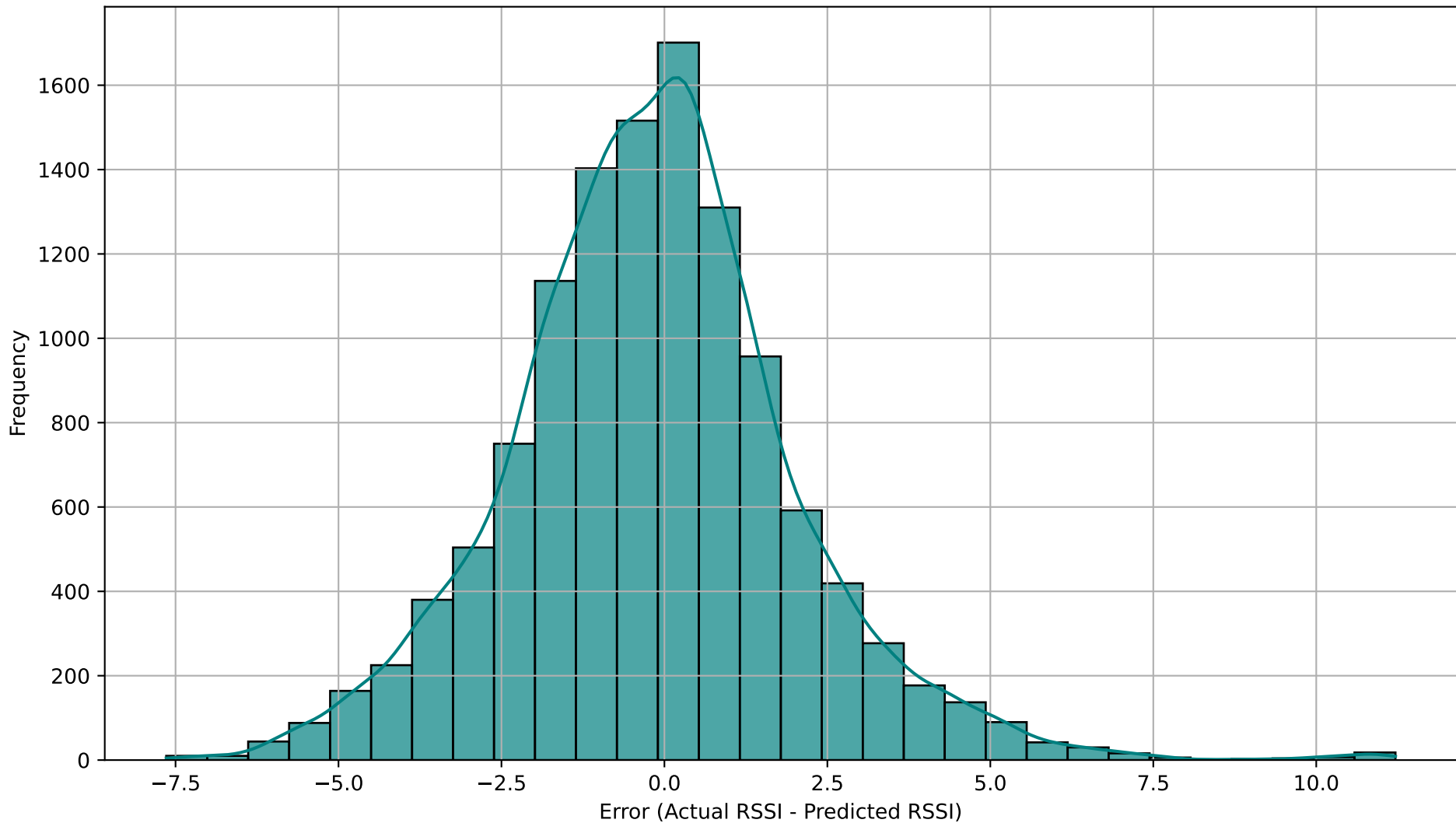
Actual vs Predicted RSSI (Last Step - step 40 $\approx$ 4.0 s)



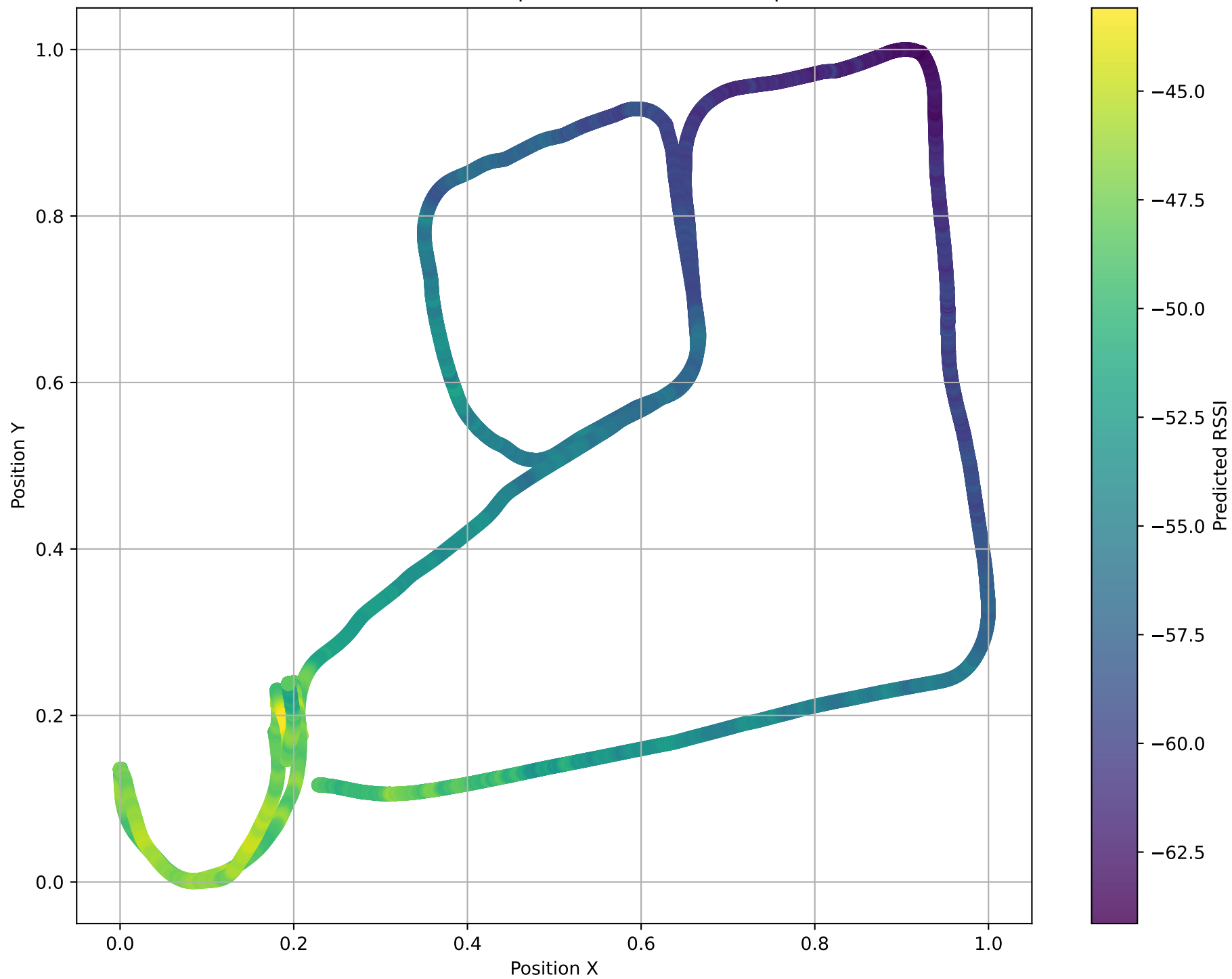
Distribution of Prediction Errors (First Step)



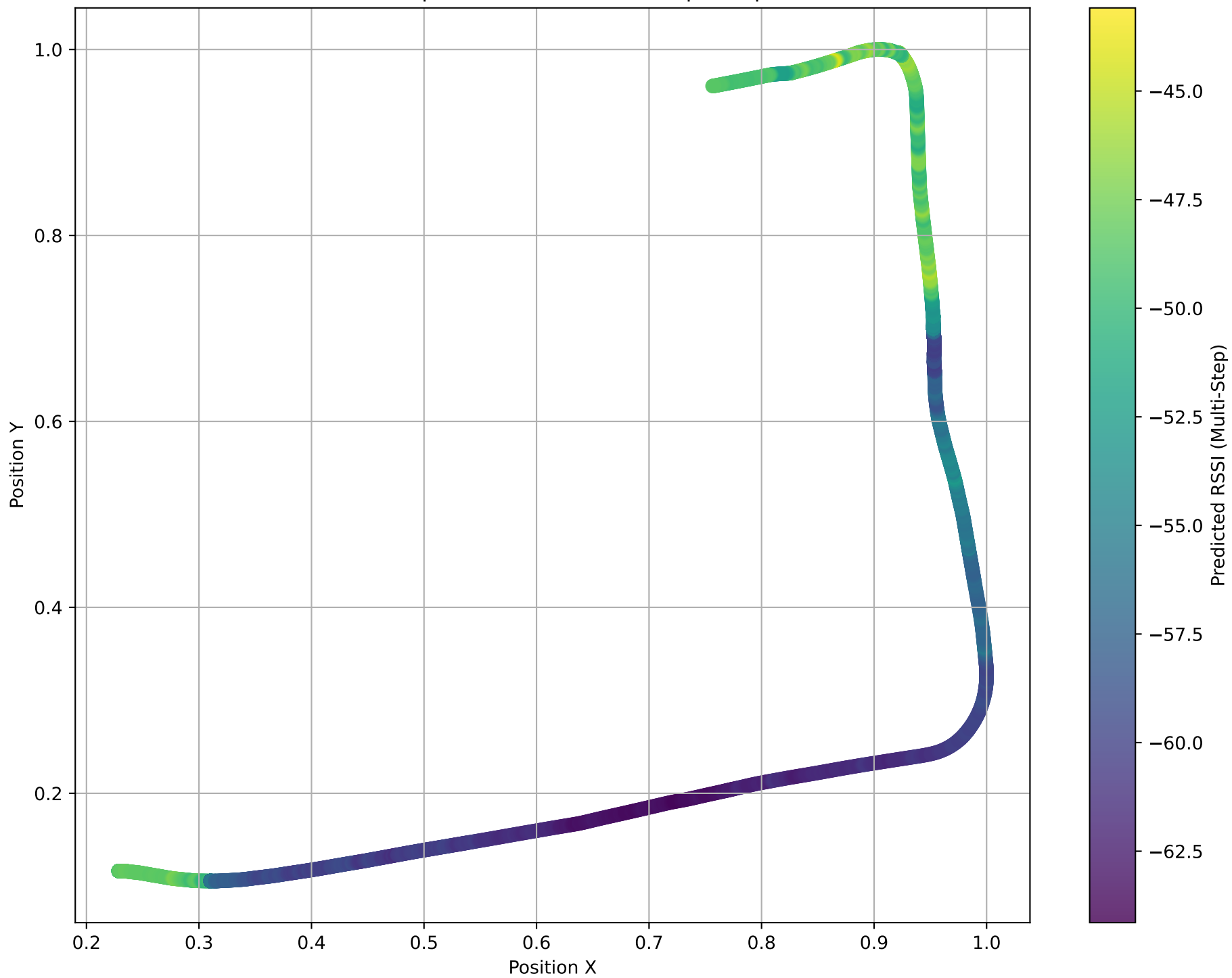
Distribution of Prediction Errors (Last Step - step 40 $\approx$ 4.0 s)



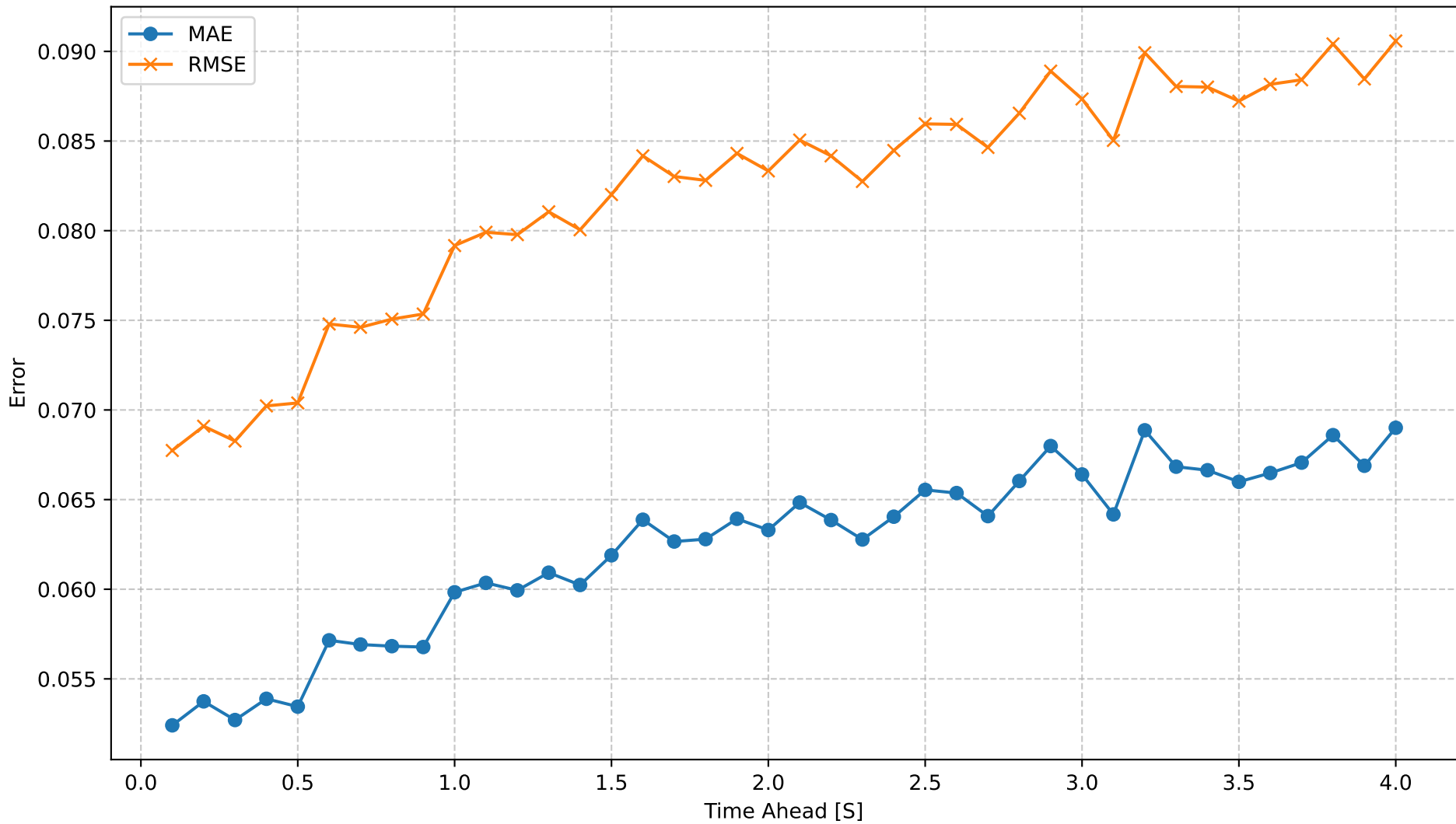
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 40 $\approx$ 4.0 s)



Multi-Step Forecast Error by Time Ahead



Resumo Automático

Dataset: Training: rssi_odom_dataset_12apr25_isolado.parquet |
Experiment: rssi_odom_dataset_11apr25_isolado.parquet
Modelo: LSTM
Forecast steps: 40

MAE / RMSE (dB)

- 1º passo : 0.05 / 0.07
- Mediana : 0.06 / 0.08
- 40º passo : 0.07 / 0.09

Crescimento do erro: 31.7% → Moderado (erro cresce 20-50 %)

3 piores passos (MAE): 40 (0.07), 32 (0.07), 38 (0.07)

Recomendação:

Use as previsões até a metade do horizonte; após isso a incerteza cresce consideravelmente.